Temporal Exchange — Engine Changelog for the CTO / Go backend

From your build `md5 80f050e2` (2026-06-14) → current HEAD `md5 bb2f8230` (2026-07-07). Feature-level, with a **Port** column = does the Go backend need a math/logic change, or is it display-only.

> **How to read this:** everything under "Core math" and "Close & settlement" changes the numbers your backend must reproduce. "Display only" rows are UI — port only if you mirror the front-end. The gate suite `engine/verify/lens_selfcheck.js` (31 checks) doubles as your acceptance tests — every anchor number below is asserted there.

1. Core math (port these first)

#	Change	Plain English	Port
1	**ITM option value rebuilt (linear smooth-paste, "PKG-ITM v2")**	Your build's in-the-money values were wrong: the value curve dipped *below* the exercised payoff from ~S/K 0.80 down, and the exercise boundary sat at 0.444K instead of the correct 0.667K (γ=2 put). Now the waiting-value power curve is welded *tangentially* (same value & slope) onto the LINEAR payoff line. Put boundary `S*=K·g/(g+1)`, call `K·(g+1)/g`, `g=m·γ`; boundary value `1/(g+1)`; past it the value IS the payoff line. Guarantee (machine-checked + hard gate): quoted value never below the exercised payoff. Test anchors: γ=2,K=$100,m=1 → S*=$66.67, boundary ⅓, ATM 0.148; m=3 → S*=$85.71, boundary 1/7, ATM 0.057.	YES
2	**Trades conserve at the TRADE POINT**	Your build applied conservation at the reserve point (a repeated spec regression). Now a trade executes where the strike ray meets the curve: `x_T=x·ρ^(w−1), y_T=y·ρ^w` (ρ = tx-ray/mode); the local pair `α_T=w·x_T, β_T=(1−w)·y_T` is conserved; `Δx=−α_T·β_T·Δy/((y_T−β_T)(y_T+Δy−β_T))`; new `w′=α_T/(x_T+Δx)`. Anchor: (x,y,w)=(10,10,½), ray θ=4, Δy=1 → Δx=−5/22, x′=215/22, **w′=11/21** exactly (not 22/43, not 6/11). Spot swaps (ρ=1) reduce exactly to your old formula — spot/arb/rebase code byte-identical. Consequence: α,β genuinely move on off-ATM trades by design.	YES
3	**Vol knob: constant slope-multiplier `m`** (if not already in your line)	Single scalar knob: lensed option-value steepness `g_loc = m·γ`, constant at every strike (m=1 = plain curve; bigger m = steeper everywhere + trade lands further out via `θ_tx=mode·(chosen/mode)^m`). Read and written through one shared helper; pool stays plain Balancer.	YES

2. Close & settlement — NEW in update-1 (2026-07-07)

#	Change	Plain English	Port
4	**Close is ONE path — a sell-back trade for every leg**	The old close had two branches: OTM legs reversed on the AMM, ITM legs were pulled off and cash-settled — a seam that made total payout JUMP ~45–87% at the OTM→ITM boundary. Now EVERY leg closes as a live reverse trade on today's curve (`tradeUpdateAt`), ITM included, no cash-settle branch. Payout is continuous across the strike (measured: crossing step 0.48× median = no jump). The trader is credited option/perp VALUE only (`L0·raw_net·carvedEquity`, raw_net = option values locked *before* the swap) — the AMM swap is pool-internal bookkeeping, never credited to a wallet.	YES
5	**Funding charged on the**	Funding weight `mark → mark − intrinsic` (intrinsic = parity arm: put `max(0,1−mode/θ)`, call `max(0,1−θ/mode)`). Funding becomes a single hump peaking at-the-money and	YES

	option-part (extrinsic), not full value	**exactly zero past the free boundary S*** (deep ITM). Reason: an ITM option is mostly the underlying, funded by the perp layer — don't double-charge. The pool-imbalance sign `±g·(S−1)/S` is unchanged. Note: on an equilibrium pool a pure spot move gives zero funding (oracle cancels); funding responds to pool disequilibrium.	

⛔ **Residual x-drain — READ before multi-party launch.** The new close does NOT round-trip pool reserves exactly (the old close did). A round trip leaves a small underlying shortfall (`Δx<0, Δy=0` exact). Verified characterization:

- **Non-extractable by construction** — the trader gets option value only; no code path credits the swap's dx/dy to any wallet (credit path byte-identical across the change).
- **Impermanent-loss-like / recovering** — if price leaves and returns, the residual recovers to a fixed small value independent of the excursion.
- **Bounded, ∝notional²** (e.g. −$29 on a test band). Harmless in a single-user sim; a small LP-vs-cycler asymmetry in a multi-party pool.

PARKED — UPDATE-2 (not implemented): a no-free-money floor + counterfactual charge-back neutralizes this and restores the exact no-free-money guarantee. **Implement it before shipping this close to a shared/multi-party pool.** Design: `specs/FIX_close_b_receipt_charge_PARKED_2026-07-03.md`.

3. Display-only (no Go port unless you mirror the UI)

#	Change	Plain English
6	Chart-2 true-value X wings + %/$ toggle	Both wings drawn OTM→ITM crossing at ATM; pool-quoted continuation solid, escrow-parity tail dashed; fraction/dollar toggle. Old peak-at-1 tent retired.
7	Vol-direction caption fix	MORE volatile asset ⇒ LOWER m (fatter wings). Your build said the opposite.
8	m clamped to [1,6]	Out-of-range m silently drove the lens out of range; now clamps with write-back.
9	Funding P/L column per position line	Portfolio shows accrued funding as a SIGNED P/L number (negative = line paid). Line P/L = base + funding×oracle. Ledger stores "trader pays" as positive — display negates it (get this sign right). Disclosure shown: displayed P/L includes accrued funding; cash at close settles ex-funding until the funding transfer layer ships.
10	Caption/label truth-ups	Invariant-Watch states the trade-point law (α/β move off-ATM by design); "% of escrow unit" → "fraction of escrow unit"; band preview animates the actual per-leg path.

4. Verification you can lean on

- **Gates** `engine/verify/lens_selfcheck.js` **— 31 hard checks** (+ `a16_atm_gate.js` 5). Every anchor number above is asserted; each check is negative-controlled (a sabotaged engine fails exactly the intended check). Use them as your Go acceptance tests.
- **CM6-v2 retired:** the old "exact round-trip / no-free-money" gate is gone (the new close is a pool-reprice close). No-free-money returns with the parked update-2 floor.
- **Formal (Lean, via Harmonic's Aristotle; artifacts on request):** value ≥ payoff, the seam weld + uniqueness, trade/rebase commutation.

5. Ruled but NOT in this handover (so nothing surprises you)

- Update-2: charge-back / no-free-money floor + LP & multi-wallet defenses (parked — needed before multi-party).

- Funding cash transfer between clubs (rate law exists; transfer layer is future).
- Read-layer smoothing (EMA-banded y) + skew rate-limit (calibration campaign, future).

Latest engine file: `engine/builds/HEAD_temporal_mvp_v28_lens.html` (md5 `bb2f8230`). Prior close retained as the revert twin `temporal_mvp_v28_lens_twocaseclose.html` (`51342574`). Generated 2026-07-07.